

Optical and Performance Comparison of Two AR Headsets

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1 Introduction into AR Systems:

Augmented Reality (AR) is a near eye display system that usually comes in the form of a headset. These headsets rely on optical systems to project images at the correct viewing angle while maintaining proper image quality. AR is a form of mixed reality (MR) which mixes forms of physical media (reality) and virtual reality (VR). AR headsets use pass-through technology that allows the user to see both the physical world and virtual elements overlaid onto the glasses [2].

1.1 Project Goals:

The purpose of this project is to perform a quantitative comparison of two commercial AR headsets using experimental optical characterization methods. The project will focus on optical performance, image quality, and user experience. This project extends the standard EP 320 course content by introducing practical measurement techniques used in electro-optical engineering and near-eye optical systems to satisfy the honors requirements.

1.2 Why it matters:

AR devices make it possible to view information and objects in a hands-free environment. This is important in fields such as medical, industrial, educational, and more. The devices can be worn indoors and

outdoors, are lightweight compared to VR, and can be combined with traditional glasses / contacts as well as work glasses for protection and practicality [4].

2 Equipment:

The following is a list of equipment that was used to perform the optical characterization of the two AR headsets:

- Smartphone Camera
- Ruler
- Desktop Computer
- Magic Leap 2 AR Headset
- XREAL Air 2 Pro AR Glasses



Figure 1: XREAL Air 2 Pro AR Glasses



Figure 2: Magic Leap 2 AR Headset

The two headsets are the Magic Leap 2 AR Headset and XREAL Air 2 Pro AR Glasses. These specific models were chosen due to the availability of AR glasses on hand at the time. A special thanks to the Embry-Riddle Extended Reality Lab for providing the

equipment and space to perform the development and testing.

3 Measurement Equations:

3.1 FOV

The horizontal and vertical field of view was experimentally. A calibrated grid pattern was displayed inside the headset, and the visible width was measured. The angular field of view was calculated using the following:

$$\theta = 2 \tan^{-1} \left(\frac{\omega}{2d} \right)$$

Results were represented as FOV_H and FOV_V . The ω in the equation is visible width and d is the estimated virtual image distance.

3.2 Angular Resolution

Angular resolution was estimated using resolution targets displayed inside the headset. The smallest distinguishable feature size was recorded and converted into angular resolution using this formula:

$$\theta_{res} = \frac{s}{d}$$

The results were converted to arcmin.

3.3 Geometric distortion

Geometric distortion was evaluated using a square grid pattern. Images were recorded through the headset optics, and distortion was estimated using the following equation:

$$Distortion = \frac{x_{measured} - x_{center}}{x_{center}} \times 100\%$$

Measurements were performed at the center and edges of the field.

3.4 Exit pupil location and eye box size

The exit pupil location and eye box size were estimated experimentally. Measurements included: Exit pupil distance, horizontal eye tolerance, vertical eye tolerance. These parameters determine viewing comfort and alignment sensitivity.

For each measurement the standard deviation was calculated using:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

4 Measurement Methods:

I used the Unity Game Engine to design, develop, and measure the FOV, angular resolution, geometric distortion, exit pupil location, and eye box size. This section will go into detail in how the scenes were programmed, changes made between headsets, difficulties that arose, and techniques applied to get the results.

I used unity to create digital scenes which were built and exported to the .APK files which can be run on the headsets, as they both run on Android. The uploading process varied slightly between the two but is irrelevant for actual measurements. The actual code will not be discussed but is available to be disclosed if interested.

4.1 FOV

Measuring the FOV on both headsets required a square grid pattern to be displayed at a fixed distance away. This is so I can visually count the number of squares before getting cut off by the glasses themselves. I was able to program this into both headsets so I would see the exact same

grid at the same distance of 1.0m no matter where I looked. From there I looked into both headsets and counted the squares vertically and horizontally. For reference, the image I saw is this blue square grid pattern:



Figure 3: Grid pattern

The dimensions is 7801 x 4001 which is way too large to be displayed fully and counted with eyes alone, so I scaled it down so that each box is 82pixels which in unity equates to 0.0246m per box at the scale I had set.

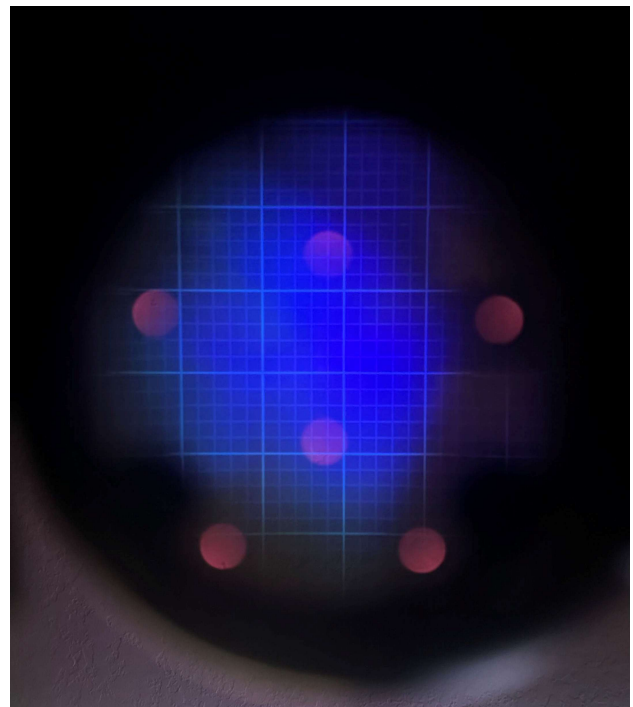


Figure 4: Grid pattern through Magic Leap

This figure shows the grid pattern as displayed in the Magic Leap 2 taken by my

smartphone camera. In addition to my visual counting using my eyes, I was able to take a screenshot using the Magic Leap connection software that I was seeing. So, I exported the picture into GIMP which is seen below:

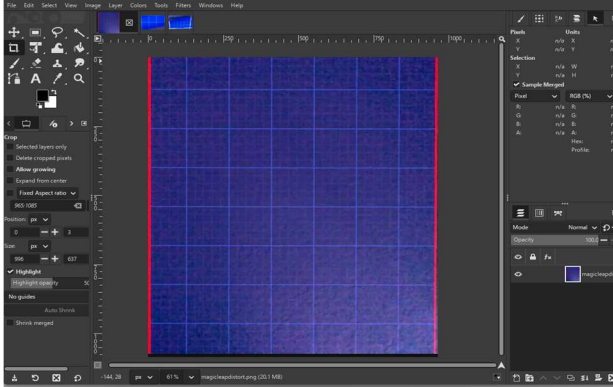


Figure 5: Grid pattern displayed in GIMP

GIMP is an image editing software that allows visual inspection of pixel values. From here I counted the same as what I saw in person, validating my results. I did the same for both headsets and got accurate results which will be covered in section 7 Results.

4.2 Angular Resolution

To find the angular resolution, I created a new scene separate from the grid pattern for both headsets. This scene has two identical cubes both set to $(0,0, \pm 0.1)$ so they are linearly separated by 0.2m. The cubes are programmed to move continuously backwards (-z) forever. I set it up this way so that when the cubes are unresolvable from one another is when I reach the minimum angle, they can be distinguished. I repeated this trial several times in different lighting to ensure I got similar results.

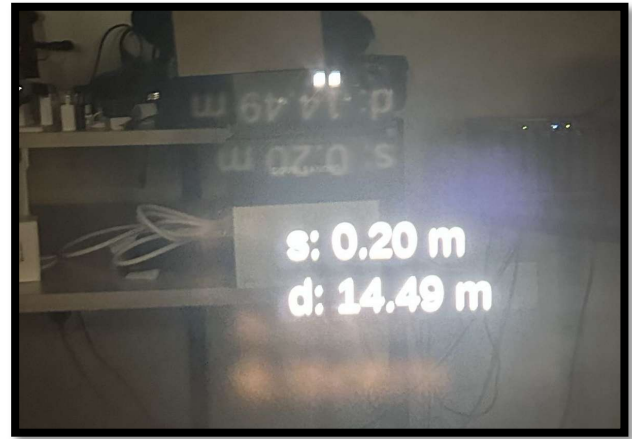


Figure 6: Angular resolution scene in XREAL

A picture taken from my smartphone shows the scene through the glasses of the boxes moving backwards in a dimly lit room. I did the same process for both headsets.

4.3 Geometric distortion

To find the geometric distortion, I used the square grid pattern as described earlier. Instead of counting the square grid visually, I had to use GIMP to calculate the pixel values as my eyes could not distinguish the difference between the center and edges of the grid. Here I simply found the center x and y coordinates, compared them to the edge coordinates, with the known pixels per box to find how much it is distorted for each headset.

4.4 Exit pupil location and eye box size

To find the exit pupil location and eye box size, I used the square grid pattern scene again. Except, I added a red boundary box with the calculated FOV being the farthest I can see visually which can be seen in Figure 5. To calculate the exit pupil distance, I started by wearing both headsets as close as possible, seeing the red border. Then I started slowly pulling the headset away from my face

horizontally (front) and stopped moving when the red borders started to disappear, or vignette. At this point I measured the gap between the lens and my eye, which is the max eye relief. I did this using a standard mm ruler and markings. Then to measure the eye box I started with the headset fully on again, red border fully in sight. Then I shifted my vision to the far left edge focusing on the border and started to slowly shift the headset horizontally (right) while keeping my head still. I stopped moving the headset when the red border started to dim/blur/disappear, then measured the movement relative to my nose. I repeated this for the vertical measurement on both headsets as well.

5 User Evaluation:

A small group of participants were asked to evaluate both headsets on a basis of clarity, comfort, ease of alignment, and eye strain. They were asked to rate each on a scale from 1 to 5, with 1 being the worst experience / headset they have worn to 5 being the best experience / headset they have worn. These results were averaged and are presented in the following table:

User Evaluations	Magic Leap 2	XREAL Air 2 Pro
Image Clarity	3.29	4.86
Comfort	2.71	4
Ease of Alignment	4	3.29
Eye Strain	5	3.71

Figure 7: User evaluations for both AR headsets averaged

6 Timeline:

The timeline for the project was completed in 5 weeks with preliminary planning and literature review, then experimental setup using the headsets, taking optical

measurements, then data analysis, and presentation and report writing. The timeline is visualized as follows:

Week	Task
1	Literature review and planning
2	Experimental setup
3	Optical measurements
4	Data analysis
5	Report preparation

Figure 8: Timeline for project completion

7 Results:

Parameter	Unit	Magic Leap 2	Magic Leap 2 o	XREAL Air 2 Pro	XREAL Air 2 Pro o
Horizontal FOV	deg	43.27	0.204	44.79	0.204
Vertical FOV	deg	52.39	0	27.64	0.408
Angular Resolution	arcmin	5.84	0.081	2.99	0.139
Vertical Distortion	%	7.08	0.204	-3.28	0.25
Horizontal Distortion	%	7.08	0.204	-0.11	0.07
Exit Pupil Distance	mm	34.67	2.054	53.33	6.236
Eye Box Horizontal	mm	6	0.816	6	0.816
Eye Box Vertical	mm	11	2.16	13.33	0.471

Figure 9: Table of compiled results

Pictured in the table above are the combined results from all the measurements and observations between the headsets. As can be seen, I was successful in measuring all required fields for both headsets.

It is important to note that the sign of the distortion denotes whether it is barrel or pincushion. This is best visualized in the figure below:

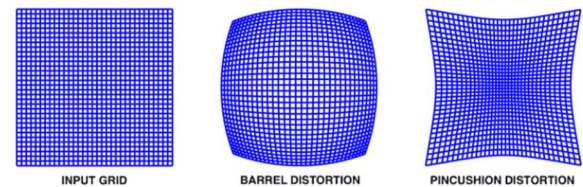


Figure 10: Distortion visualization [3]

If the distortion is negative, as is by the XREAL, then it is barrel distortion. Whereas if positive, like the Magic Leap, then it is pincushion. This effect was not extremely

visible during regular use, but with the grid applied it can be seen more evidently.

The actual data from the manufacturers' websites on all the observations are listed below in the following table:

Parameter	Unit	Magic Leap 2	XREAL Air 2 Pro
Horizontal FOV	deg	44.6	46
Vertical FOV	deg	53.6	27
Angular Resolution	arcmin	1.875	1.22
Vertical Distortion	%	NA	NA
Horizontal Distortion	%	NA	NA
Exit Pupil Distance	mm	15	20
Eye Box Horizontal	mm	20	34.4
Eye Box Vertical	mm	9	19

Figure 11: Manufacturer specs [1, 5, 6]

As it can be seen, all but the vertical and horizontal distortion was made publicly available.

Comparing between my measurements and the actual OEM specifications. I can say that my methods were greatly successful, this can be quantified by the percent error between the two as follows:

$$\%Error = 100 * \left| \frac{Experimental - Actual}{Actual} \right|$$

Where the % Error is shown in this table below:

Parameter	%Error Magic Leap 2	%Error XREAL Air 2 Pro
Horizontal FOV	3.073723134	2.70149587
Vertical FOV	2.309601069	-2.315484805
Angular Resolution	-67.89383562	-59.19732441
Vertical Distortion	NA	NA
Horizontal Distortion	NA	NA
Exit Pupil Distance	-56.73492933	-62.4976561
Eye Box Horizontal	233.3333333	473.3333333
Eye Box Vertical	-18.18181818	42.53563391

Figure 12: %Error between Experimental and Actual

The most successful measurements were the horizontal and vertical FOV as they both stayed in the range of 5% error to the actual specified FOV. This shows that the experimental method of measurement I used

is highly accurate and can be extended to any headset.

However, the angular resolution for both headsets was over 50% which is not ideal. This is likely due to the fact that the measurement tactic used was highly variable between trials and varies from person to person compared to the static FOV for a headset. There is not much I would have done differently, except complete more trials and test with different people's eyes for diversity.

The exit pupil distance suffered the same fate as being over 50% error. This was likely due to the lack of a more accurate measurement tactic than a ruler and visually inspecting the distance. A better setup would have been to use a tripod to hold the glasses and a smartphone camera to focus on the image better than using my eyes and visually inspecting. Additionally, I would complete more trials and use more people as data. The same can be said for the horizontal eye box measurements as they were significantly higher than 100% error, whereas the vertical eye box measurements were less than 50% but by no means ideal.

8 Outcomes & Takeaways:

To conclude, I would say that this project was successful, in that I was able to measure all aspects I set out for both headsets. I employed scientific measurement methods as well as programming and 3D visualization techniques. The most accurate result was the FOV measurements using the displayed grid. While other measurements showed a high % error, I have learned about the importance of measuring optical systems. After completion

of this project, I have gained skills in applying optical engineering to concepts and to real electro-optical systems such as augmented reality glasses. I have gained a deeper understanding of near eye optical systems, evaluating engineering trade-offs between optical designs, performing quantitative optical measurements, as well as gathering experimental results and presenting in professional technical format.

9 References:

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Special Thanks to Embry-Riddle Extended Reality Lab for equipment.

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